

IEA CCS Technology Roadmap

Deployment Strategies, Cost Trajectories, and Policy Framework for CCS to 2050

Reading 2 · Module 2 and Module 5 · Meer Foundation EduSuTo

Course: Climate Change: Carbon Capture and Storage

About This Reading

The International Energy Agency (IEA) has been the world's foremost authority on energy data and analysis since its founding in 1974. Its Net Zero by 2050 roadmap (2021) and subsequent CCS-specific analyses provide the most widely used framework for understanding how CCS fits into the global energy transition. This reading synthesises IEA's CCS analysis, translates its findings for course learners, and adds India-specific context.

IEA Net Zero 2050: The Role of CCS

The IEA's Net Zero by 2050 scenario (NZE) is one of the most influential decarbonisation roadmaps. It analyses what must happen across all energy sectors for the world to reach net-zero emissions by 2050.

CCS in the NZE Scenario:

Year	CCS Deployment (Gt/yr)	Key Sectors	Required Investment (USD/yr)
2022 (actual)	~0.05	Industrial (majority)	< \$5 billion
2030 (NZE target)	1.0	Power, Industry, H■	~\$100 billion
2040 (NZE target)	3.5	Industry, BECCS, H■	~\$200 billion
2050 (NZE target)	6.8	All sectors	~\$250 billion

The IEA calculates that the cost of not deploying CCS — having to achieve equivalent decarbonisation through other means — would increase the total global energy system cost by USD 4 trillion cumulatively to 2050.

The Hard-to-Abate Sectors: Where CCS Is Most Critical

Sector	CO ₂ Emissions (2022)	CCS Role	Cost Range (\$/tCO ₂)
Cement	~2.5 Gt/yr	Only viable pathway for process emissions	\$80–120
Steel	~2.6 Gt/yr	Complement to H ₂ -DRI for blast furnaces	\$60–100
Chemicals	~1.8 Gt/yr	Blue H ₂ , process capture	\$50–90
Refining	~1.0 Gt/yr	Hydrogen production, process CCS	\$40–80
Coal power	~10 Gt/yr	Retrofit and new-build; declining role	\$70–120
Gas power	~5 Gt/yr	Firm low-carbon power with CCS	\$60–100
Waste-to-energy	~0.5 Gt/yr	Net-negative (biogenic carbon)	\$80–140

Technology Readiness and Cost Trajectories

The IEA classifies CCS technologies by maturity. Understanding where each technology sits on the cost curve is essential for project developers:

Technology	IEA Maturity Level	Current Cost (\$/tCO ₂)	2030 Projection	2040 Projection
Post-combustion (amine)	Commercial	\$60–120	\$50–90	\$40–70
Pre-combustion (SMR+CCS)	Commercial	\$50–90	\$40–70	\$35–60
Oxyfuel	Demonstration	\$80–140	\$60–100	\$45–80
Direct Air Capture	Early commercial	\$250–600	\$150–300	\$100–200
BECCS	Pilot/Demonstration	\$100–200	\$80–150	\$60–120
Offshore geological storage	Commercial	\$15–30	\$12–25	\$10–20

Regional CCS Deployment by 2030: IEA Analysis

Region	2030 CCS Target (Mt/yr)	Primary Driver	Key Projects
North America	250+	Policy (45Q), market demand	Stratos DAC, Project Bison, Quest
Europe	150+	EU ETS carbon price, regulation	Northern Lights, Porthos, HyNet
China	50+	Pilot expansion, industry	Multiple industrial pilots
Middle East	50+	Oil and gas operations	ADNOC, Saudi Aramco
Australia	25+	Export-oriented H ₂	CarbonNet, Gorgon
Japan/Korea	25+	Net-zero commitments	Tomakomai, K-CCUS
India	5–10	Emerging pilot stage	ONGC, IOC pilot feasibilities

What This Means for Project Developers and Policymakers

The IEA analysis carries several critical implications for those developing or evaluating CCS projects:

- 1. Carbon pricing is the primary driver.** In regions with a carbon price above ~\$50/tCO₂, CCS becomes economically viable for many applications without subsidy.
- 2. Shared infrastructure reduces costs by 30–50%.** Hub-and-spoke models (like Northern Lights) dramatically improve project economics for smaller emitters.
- 3. Long-term contracts reduce risk.** Long-term CO₂ offtake agreements and government contracts-for-difference are essential for securing investment.
- 4. Storage characterisation must start now.** Geological assessment takes 5–10 years. Regions that start now (India, Southeast Asia, Africa) can deploy by 2035–2040.
- 5. Learning rates are significant.** The IEA projects 15–25% cost reductions for every doubling of deployed CCS capacity — similar to the learning curve seen in solar PV.